

Gut Microbiome Characterization and Machine Learning Classification of Type 2 Diabetes in Adults in an Urban African Setting

ENEZA Residential Training (Cohort 3)

7th August 2026

- **Project 8**
- Samuel Mwasya
- Linnytiffan Moreka
- Geoffrey Katana
- Samson Otiso

Introduction

- Gut microbiome plays a critical role in host nutrition, metabolism and immune function.
- Its composition is influenced by factors such as diet, geography, age, seasonality, lifestyle.
- Dysbiosis has been associated with several diseases, including Type 2 Diabetes, obesity, cancer, and atherosclerosis.
- These observations suggest that microbial profiles are population-specific, underscoring the need to study diverse populations.

Research Gap

- Numerous studies from Europe, North America, China, and animal models have reported associations between the gut microbiome and Type 2 Diabetes (T2D).
- In contrast, few studies have investigated gut microbial composition and its relationship with disease outcomes in African populations.
- Limited studies have examined the association between the gut microbiome and T2D among urban Africans.
- Given that gut microbiome composition varies across populations, findings from other populations require evaluation and validation in African populations.

Research Questions

- Does gut microbial composition differ between individuals with Type 2 Diabetes and healthy controls?
- Can microbial taxonomic abundance profiles accurately distinguish individuals with Type 2 Diabetes from healthy controls?

Objectives

- Characterize alpha and beta diversity patterns between individuals with Type 2 Diabetes (T2D) and healthy controls.
- Compare gut microbial composition between T2D and healthy controls.
- Identify the microbial taxa that best discriminate between T2D and healthy controls and assess their biological plausibility.
- Develop and evaluate a classifier that predicts disease status (T2D or healthy) from taxonomic abundance profiles.

Methodology

Study: PRJNA607849

Study year: 2019

Study type: Case-control

Location: Ibadan, Nigeria

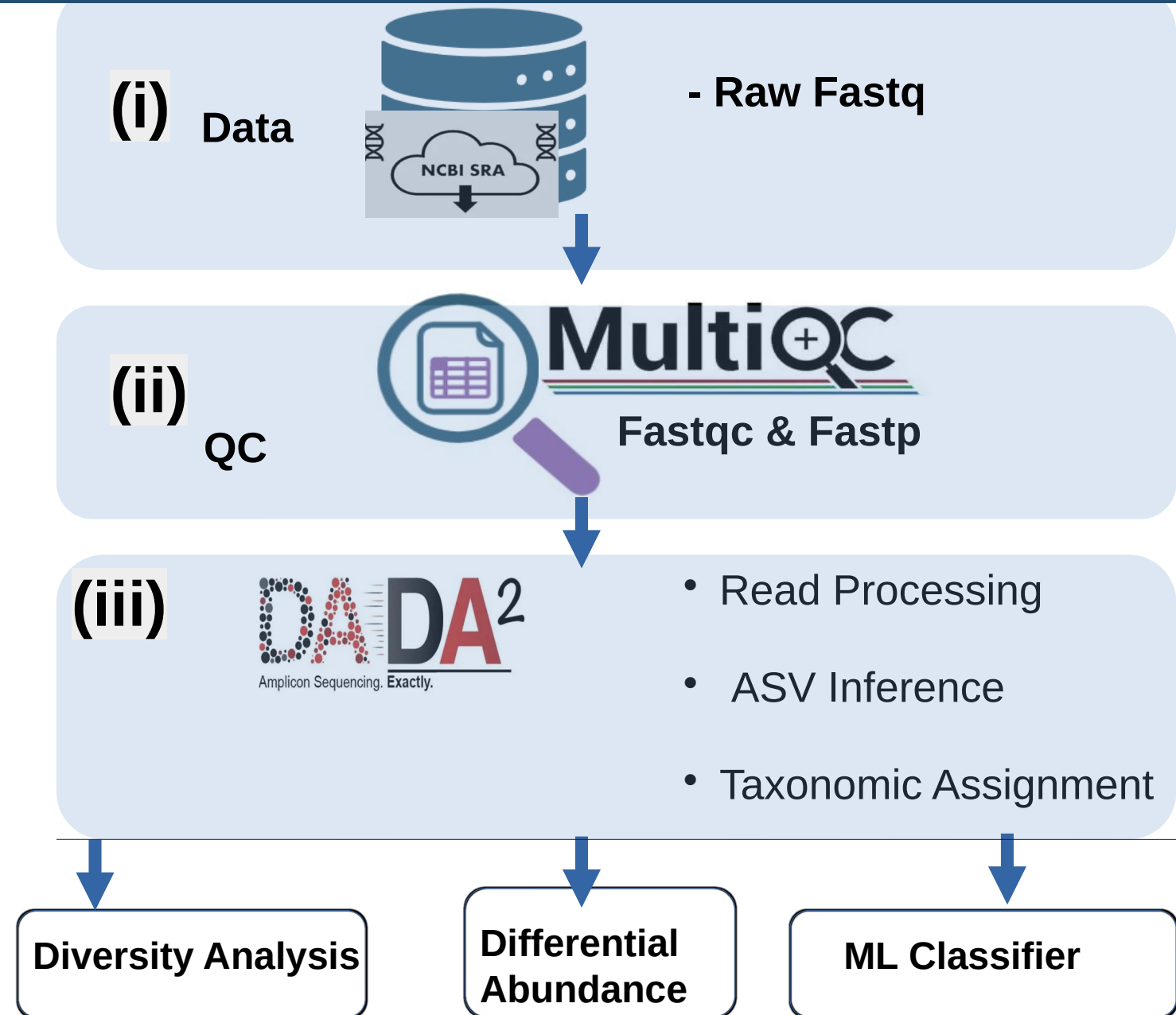
Sample data: 131 samples

Type 2 Diabetes: 66 samples

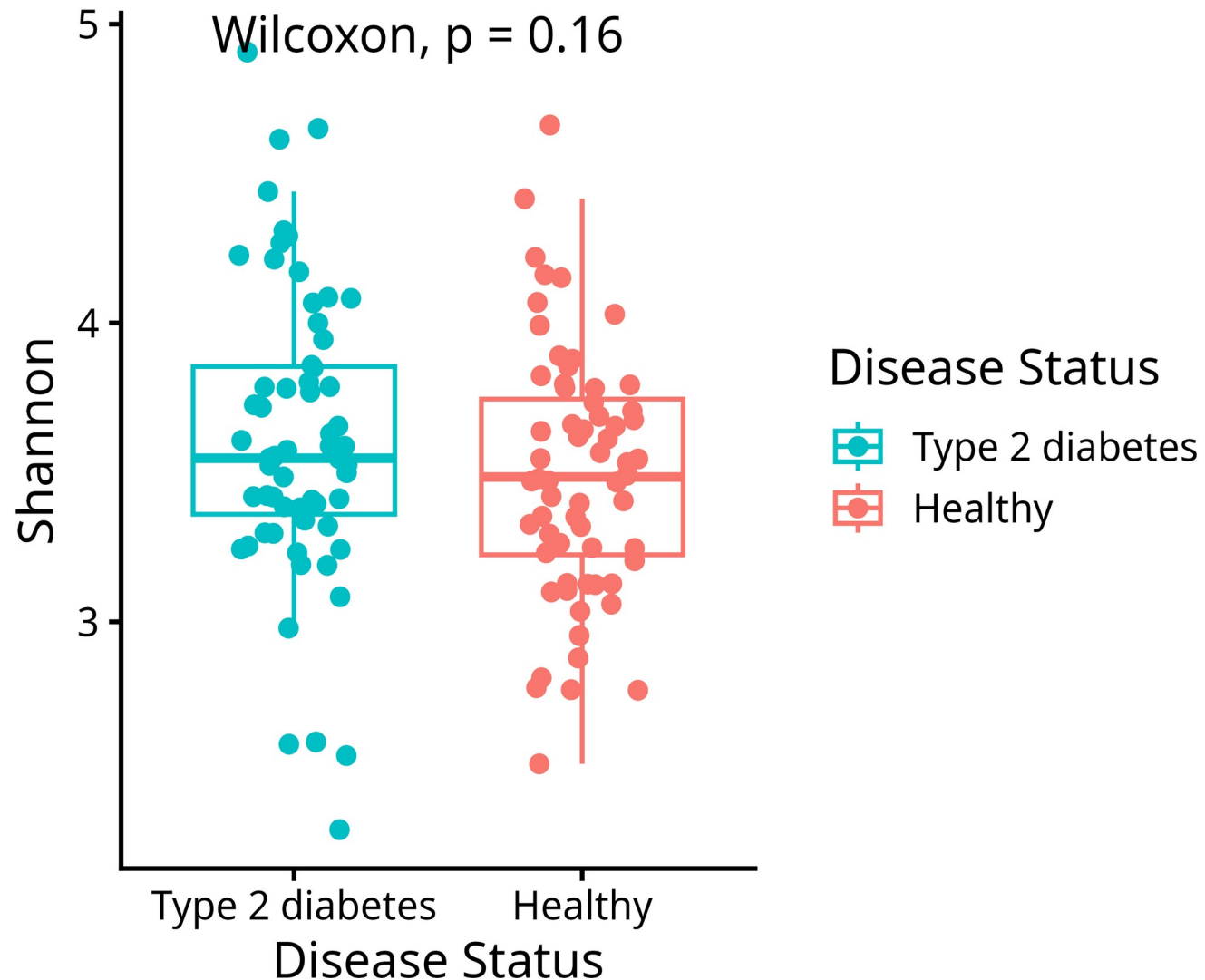
Healthy controls: 65 samples

Data source:

Publicly available 16S rRNA
sequencing fastq files from
NCBI SRA DB



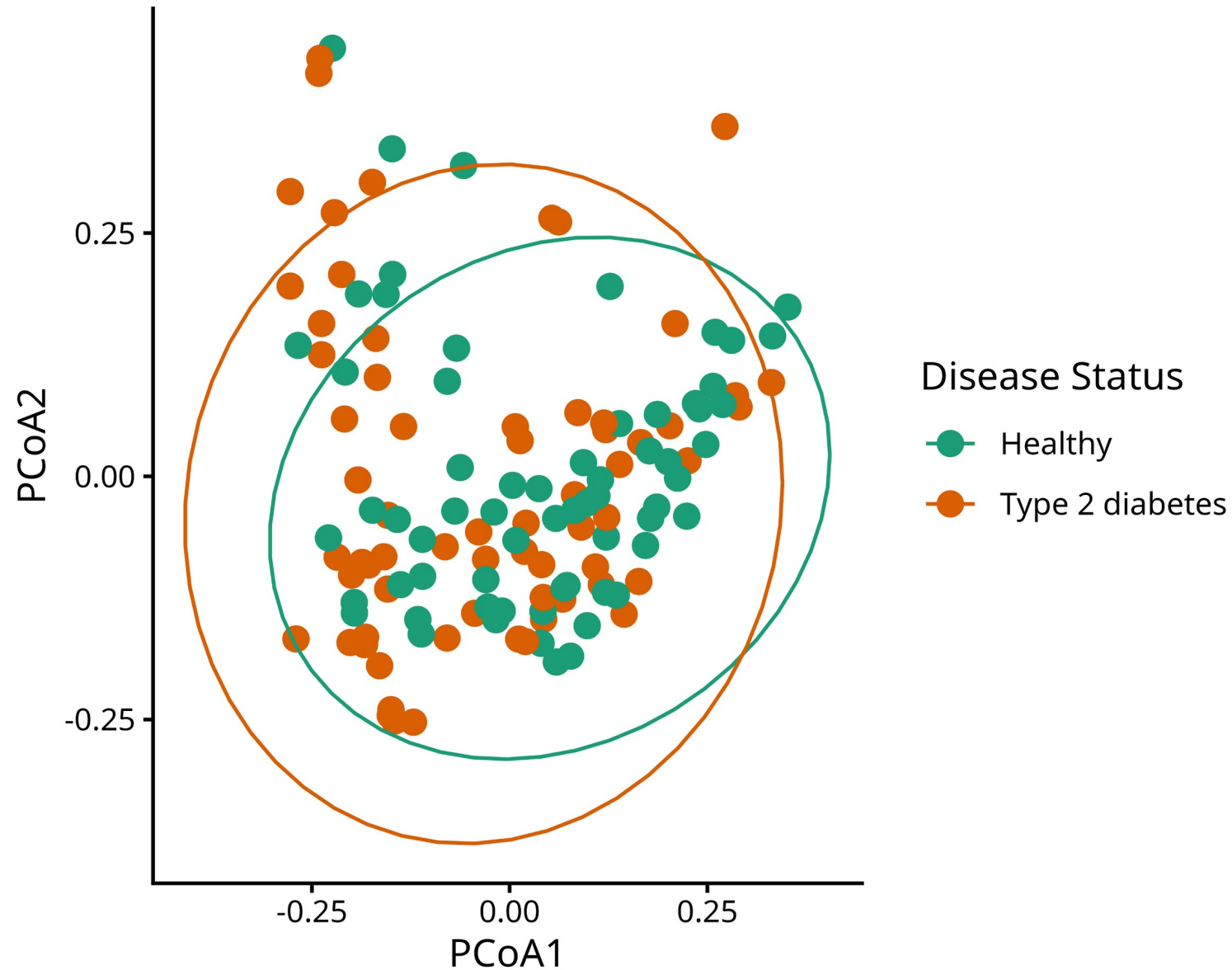
Alpha diversity within the samples



Median (IQR)

- **Healthy** - **3.496 (3.249 - 3.771)**
- **T2D** - **3.566 (3.346 - 3.879)**
- There was no significant difference in Shannon diversity between the groups.

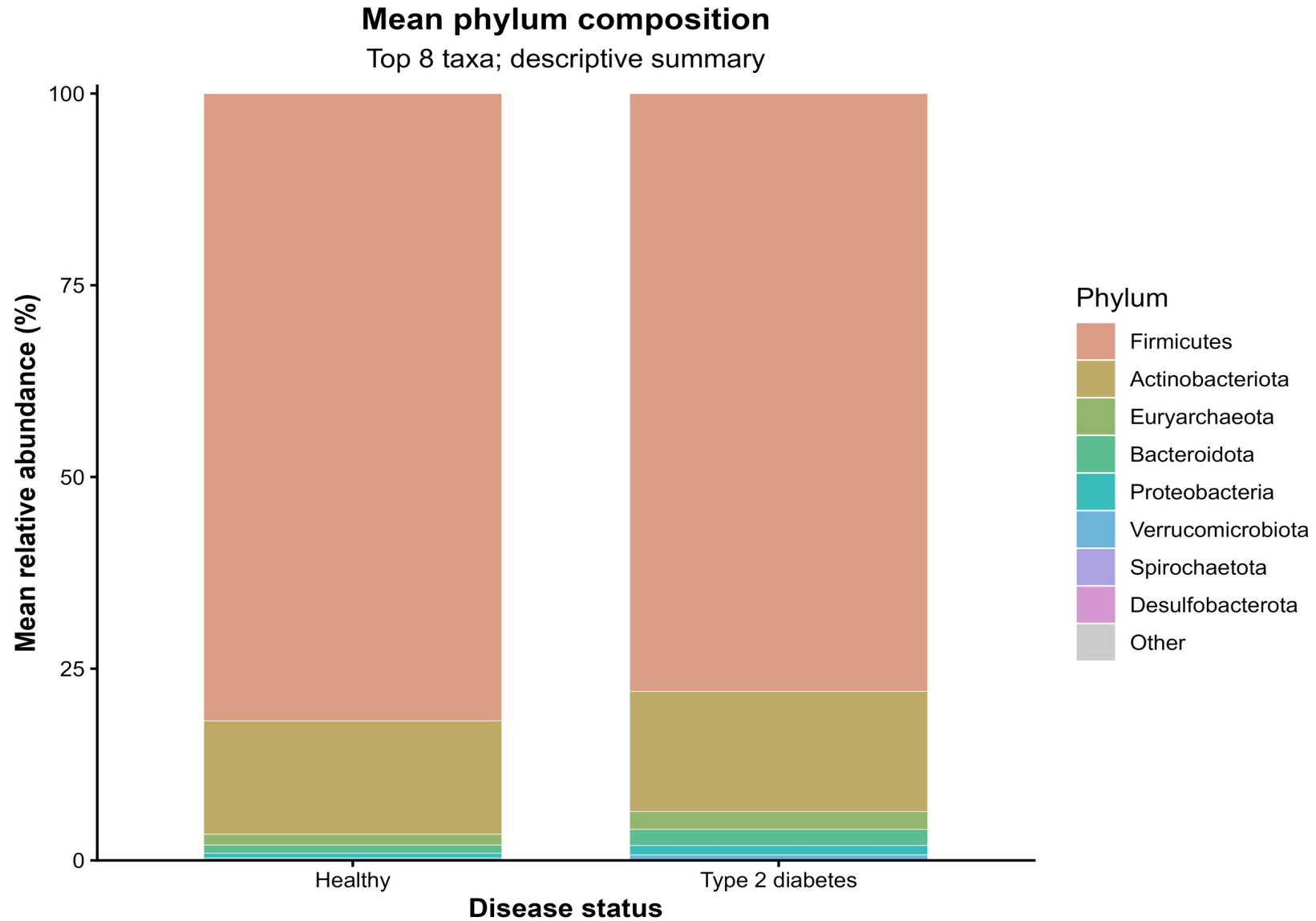
Beta diversity patterns between T2D and healthy controls



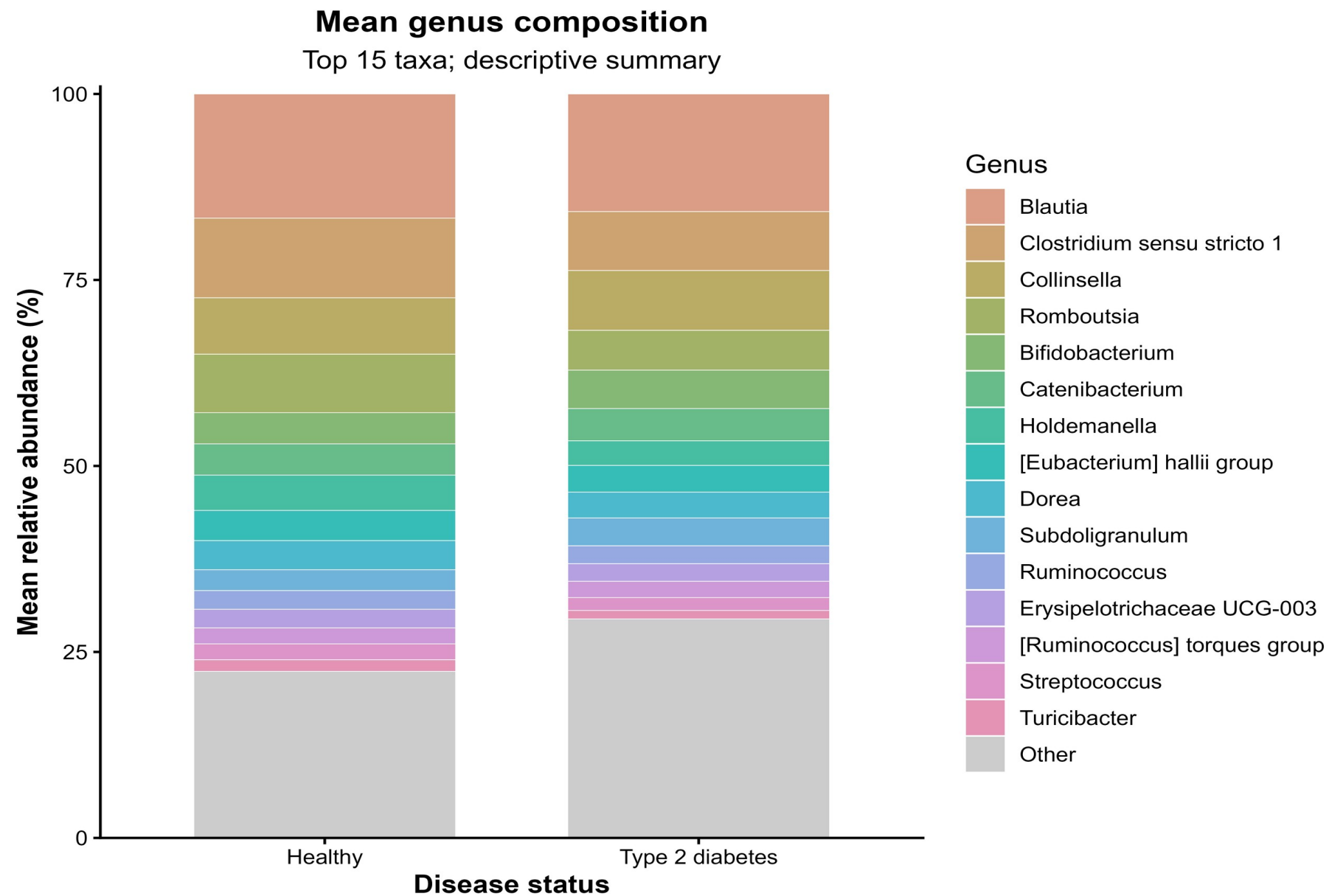
PERMANOVA

- **$R^2 = 0.013$**
- **$F = 1.65$**
- **$P = 0.026$**

Microbial composition (phylum) between T2D and healthy controls



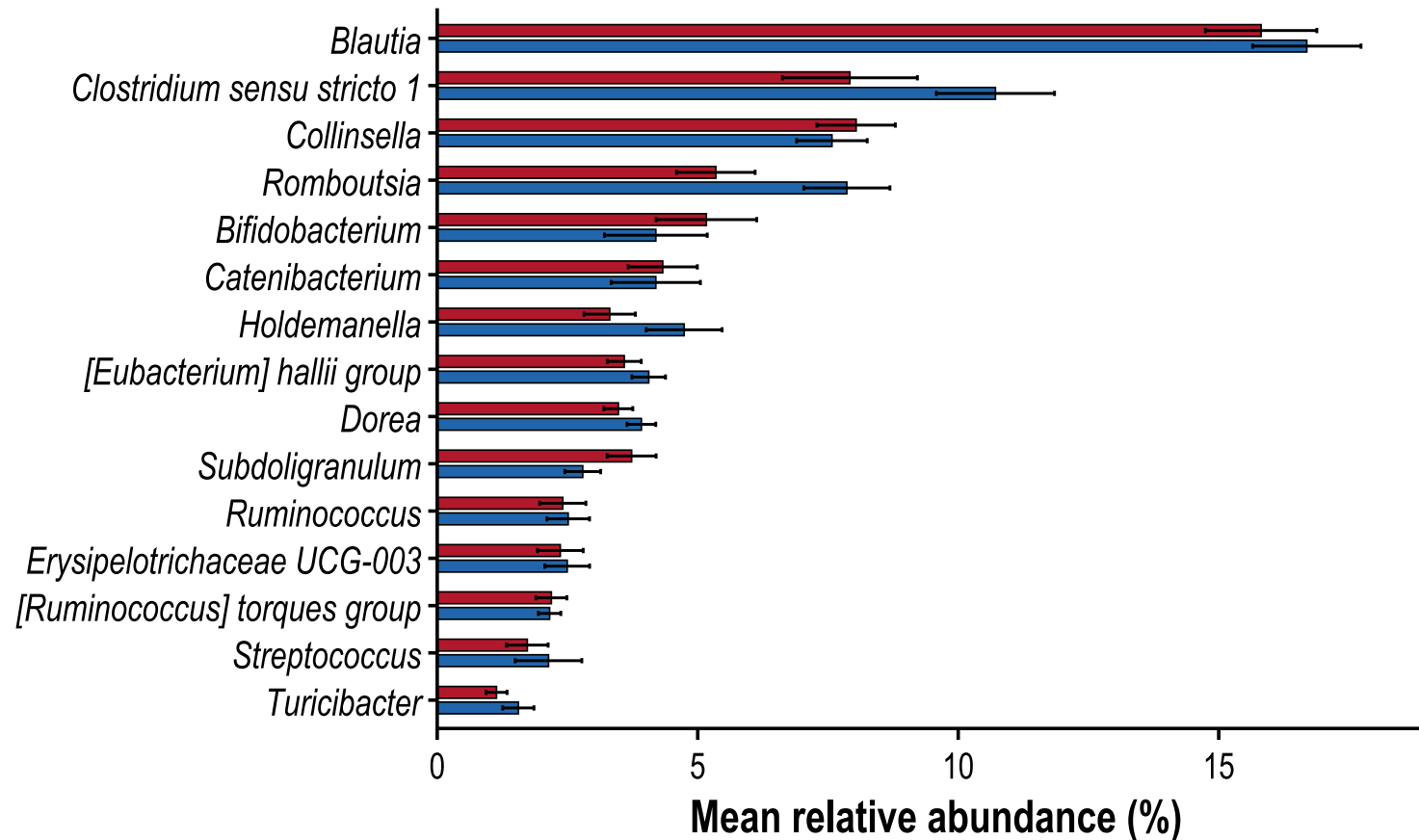
Microbial composition (genus) between T2D and healthy controls



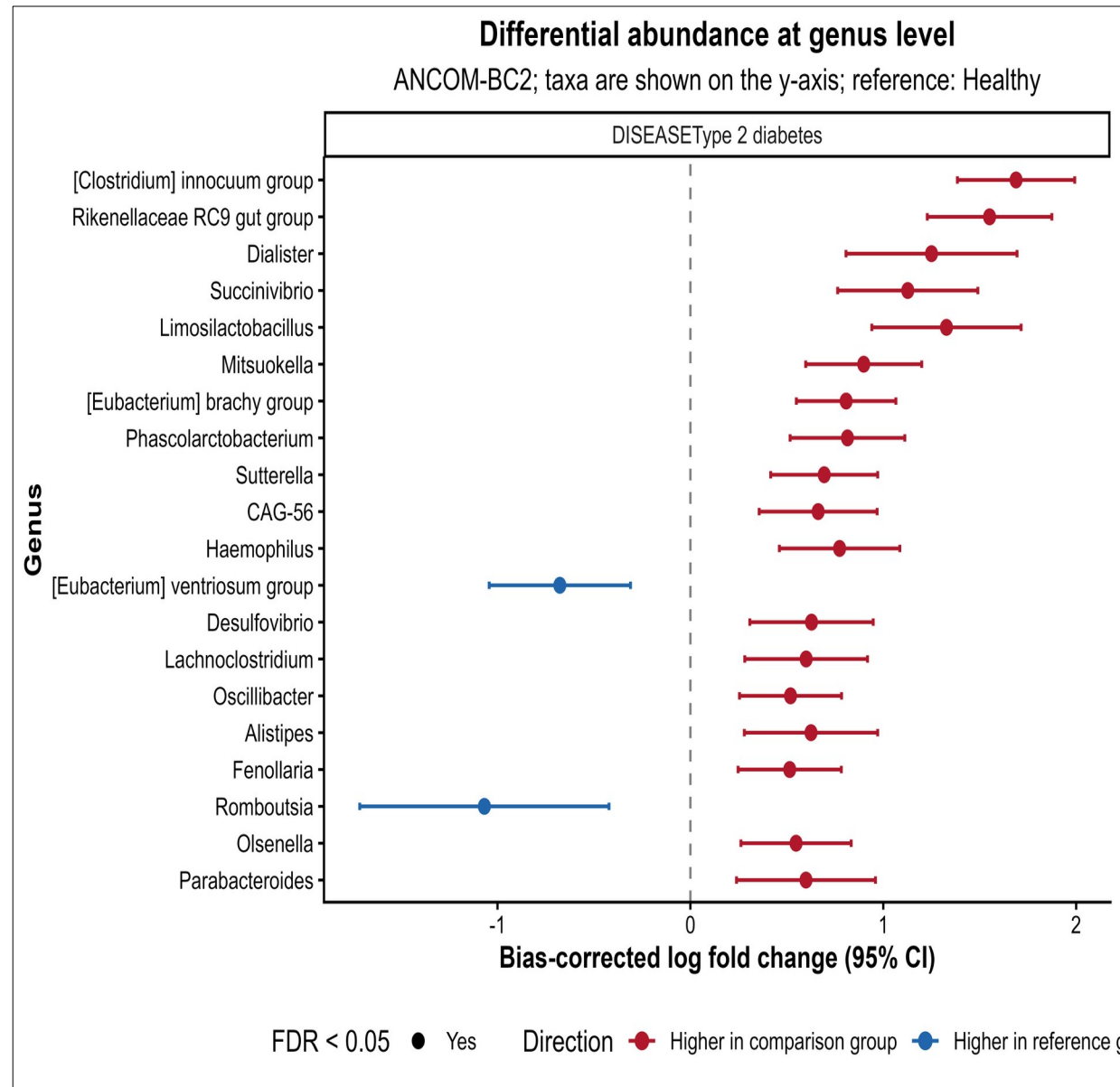
Top 15 most abundant genera by disease status

Bars show group means; error bars show ± 1 standard error

Disease status Healthy Type 2 diabetes



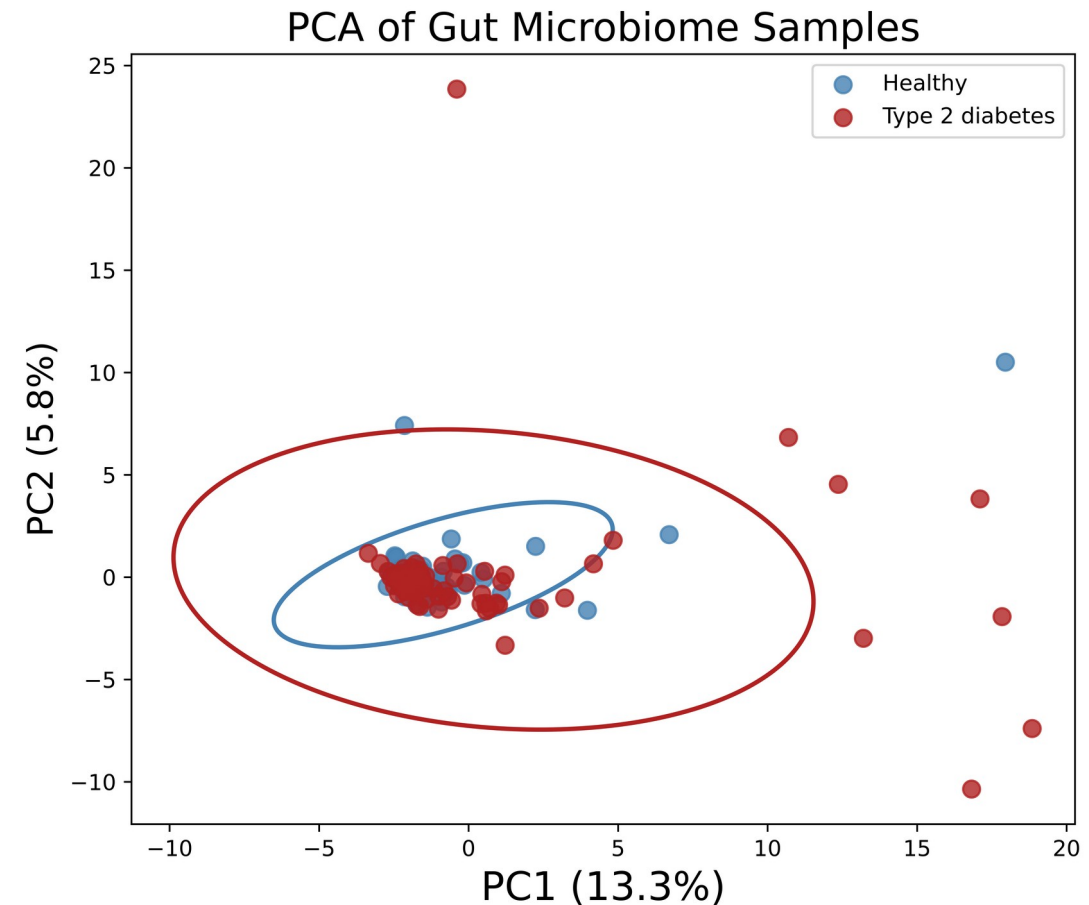
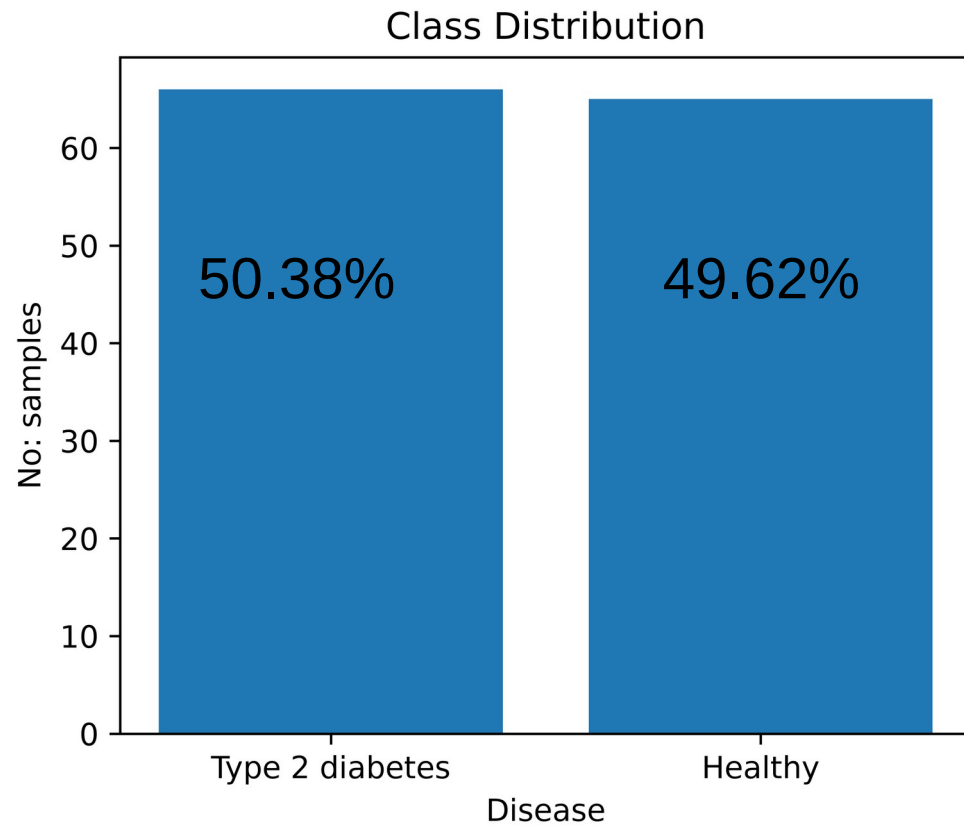
Best discriminative taxa between T2D and healthy controls



Machine Learning classifier for T2D

Baseline:

- 131 samples = 316 features
- zero missing values across all features



Noise Reduction

Start: 316 microbial taxa



0% Variance filter
`VarianceThreshold(0)`

`Selector = VarianceThreshold(Threshold = 0)`

- Removes biologically inactive or constant features.
- Features with zero variance across all samples are discarded.

`Keep = prevalence >= 0.10 * len(x)`

- Eliminate rare genera found in fewer than 10% of the samples.
- Prevent model overfitting on outlier data.



10% Prevalence filter

`x > 0`

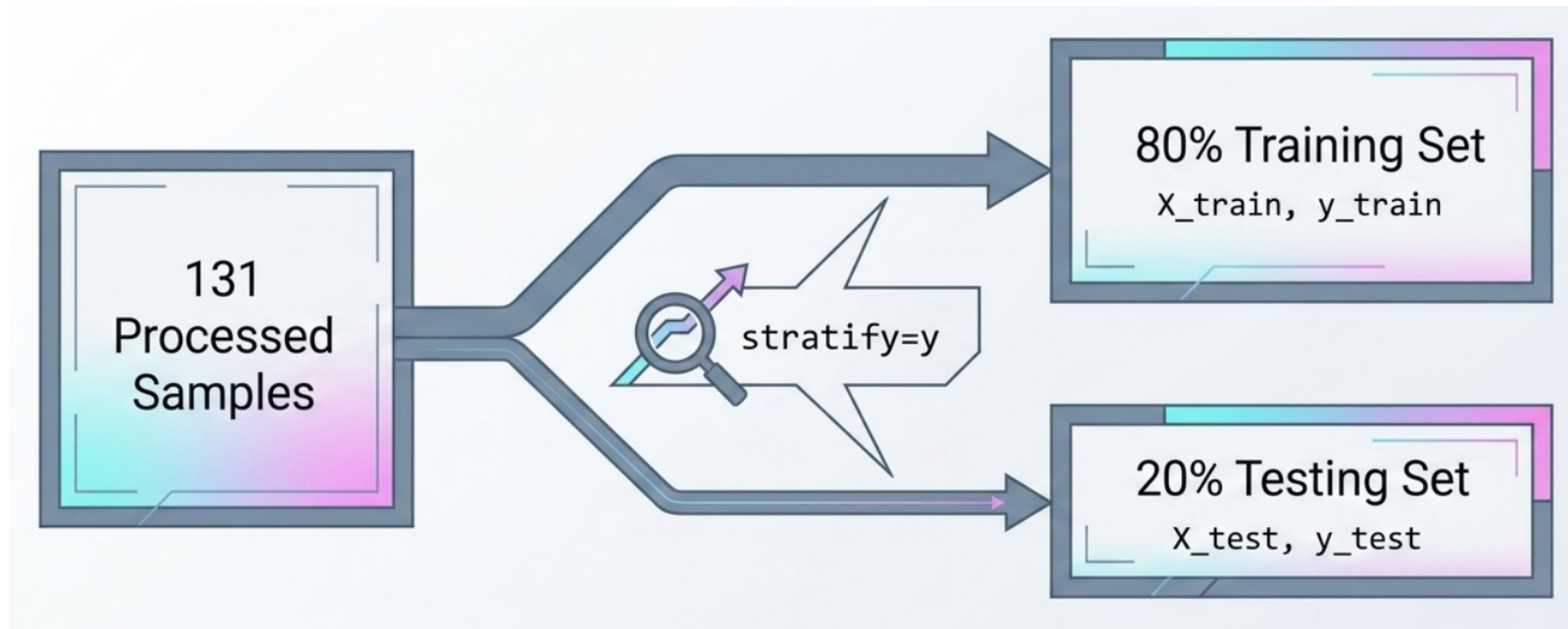
142 microbial taxa

- Cleaned high-signal feature set.
- Ready for splitting

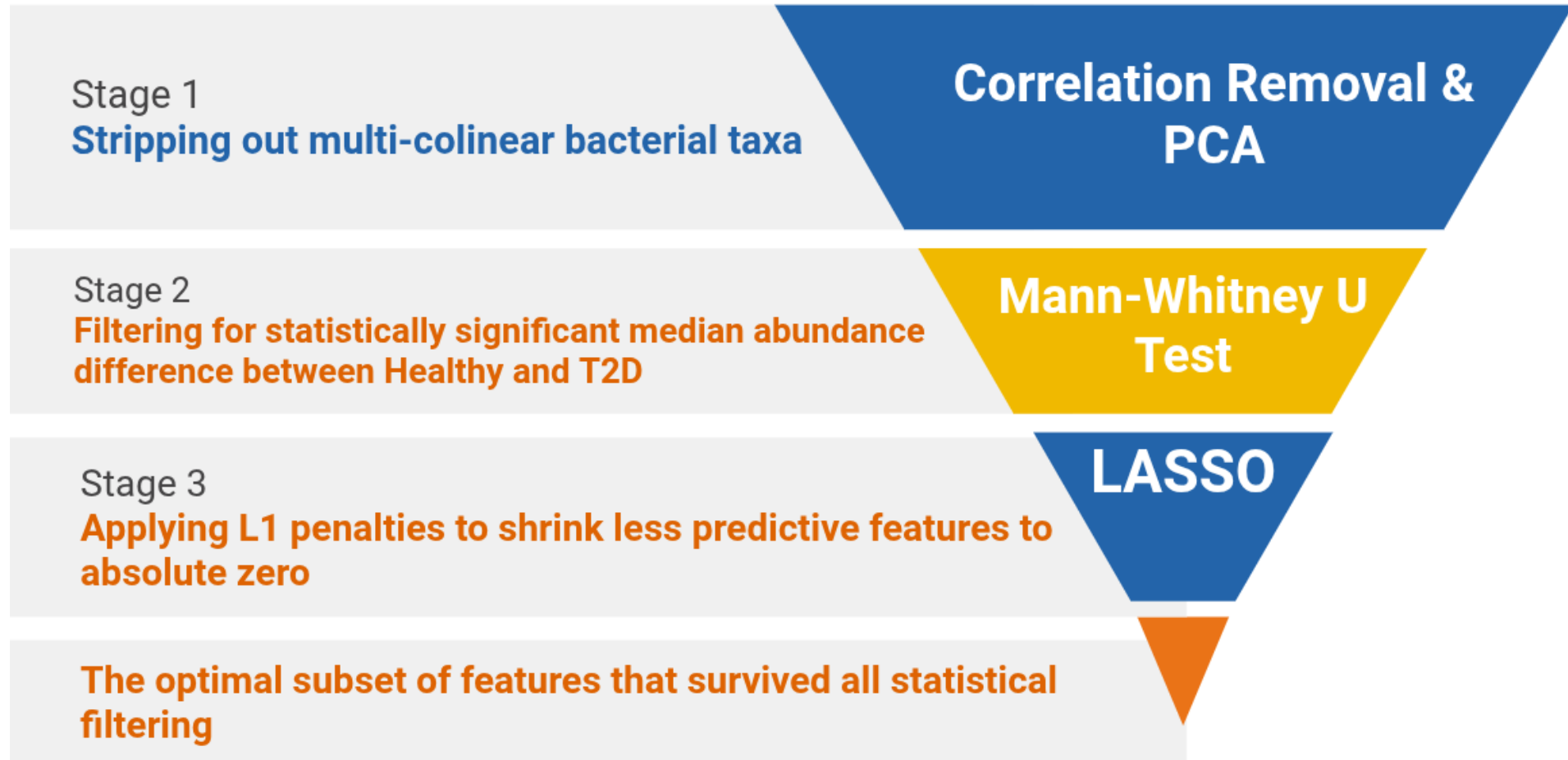
Model isolation/splitting

A 80/20 isolation:

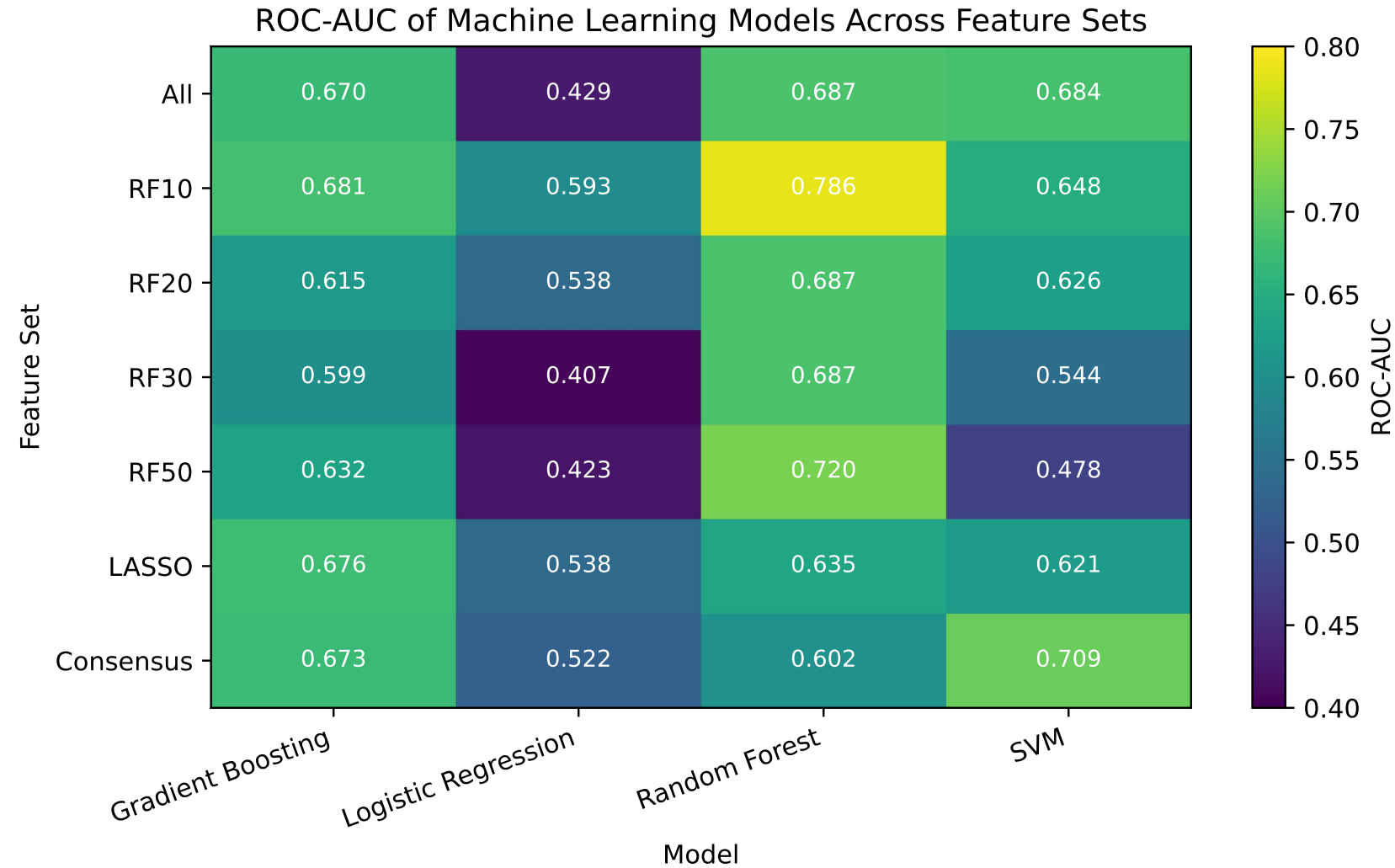
- To ensure the final model is evaluated on unseen biological profiles.
- Stratification to enforce a 50/50 healthy-to-diabetic ratio across sets.



Feature selection



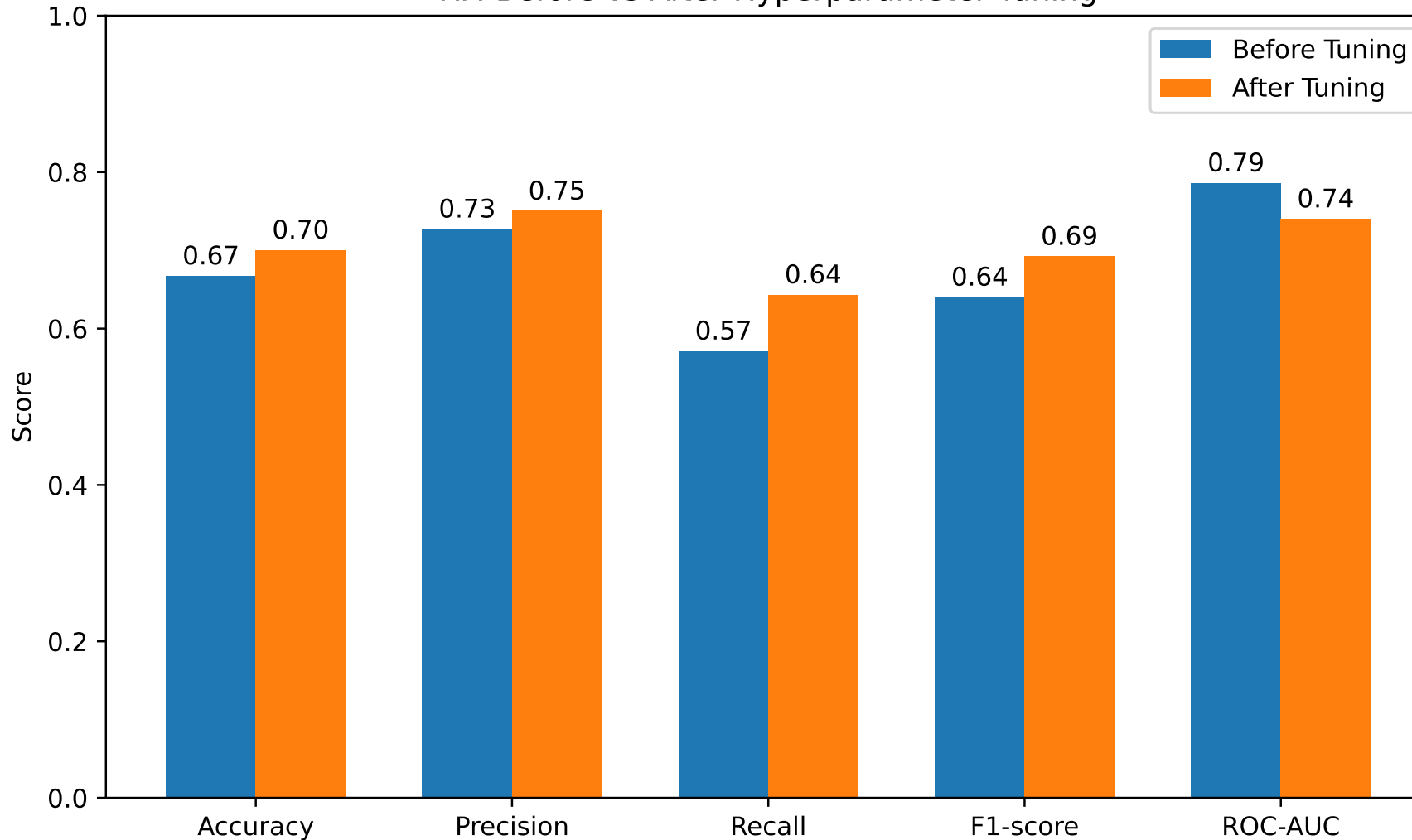
Model selection



- The Random Forest model (RF10) best with peak ROC-AUC of 0.786.

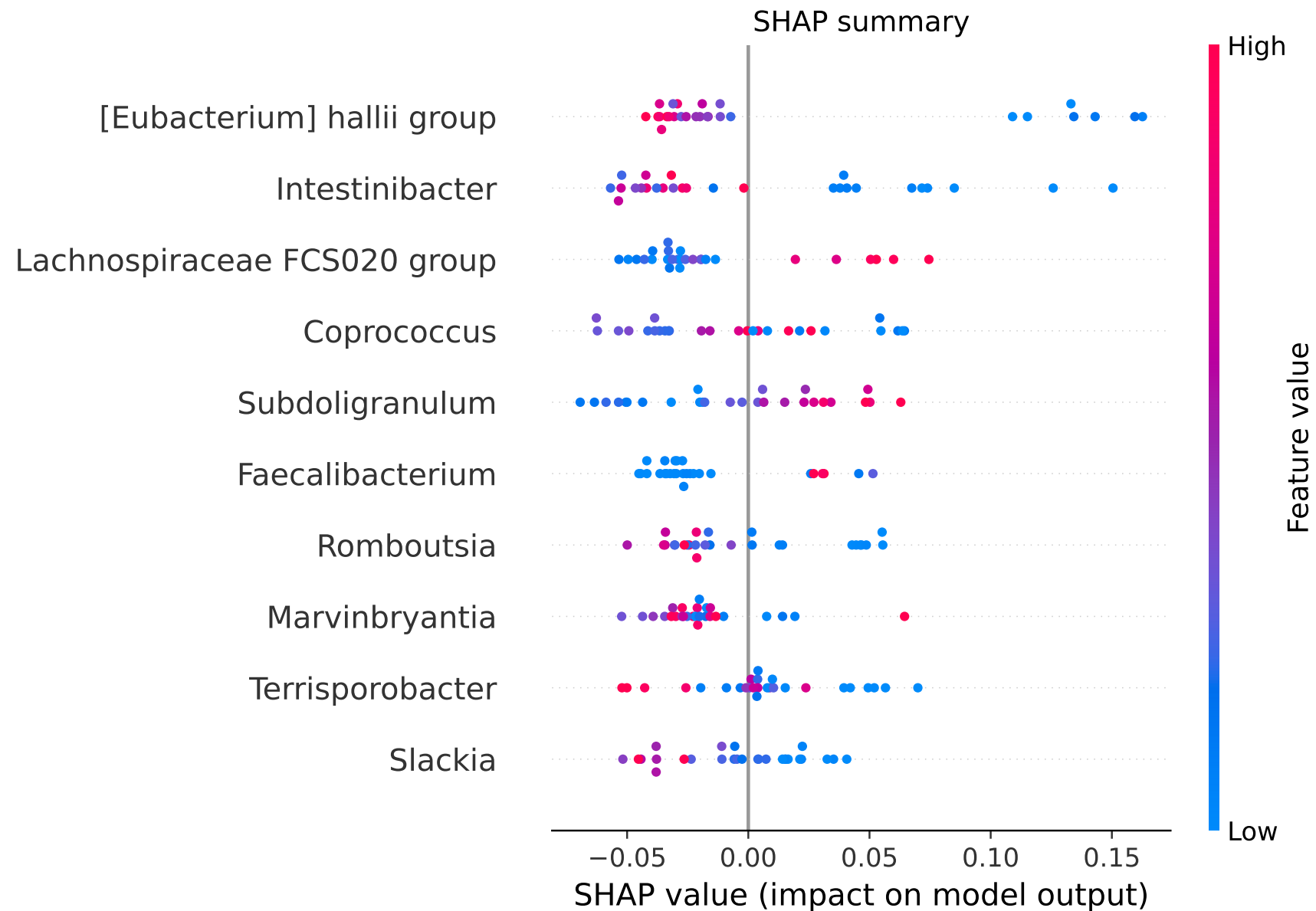
Model tuning

RF: Before vs After Hyperparameter Tuning



- Tuning the Random Forest model using, StratifiedKFold & GridSearchCV.
- Accuracy (0.70) and Recall (0.64)

Model explanation



Model packaging and deployment

RANDOM FOREST ENGINE

t2d_random_forest.pkl

The binary serialized final model containing all trained decision trees and weights.

```
joblib.dump(  
    final_rf,  
    "../model/t2d_random_forest.pkl"  
)
```



model_features.json

The strict, ordered list of the consensus 'RF10' bacterial taxa required for inference.

```
#save the feature list  
with open("../model/model_features.json", "w") as f:  
    json.dump(features, f, indent=4)
```



model_metadata.json

Operational parameters (max_depth: 5, n_estimators: 100) and training provenance for version control.

```
#save the model metadata  
metadata = {  
    "Model": "Random Forest",  
    "Disease": "Type 2 Diabetes",  
    "Features": len(features),  
    "Training_samples": len(X),  
    "n_estimators": 100,  
    "max_depth": 5,  
    "max_features": "sqrt",  
}
```



Challenges & Solutions

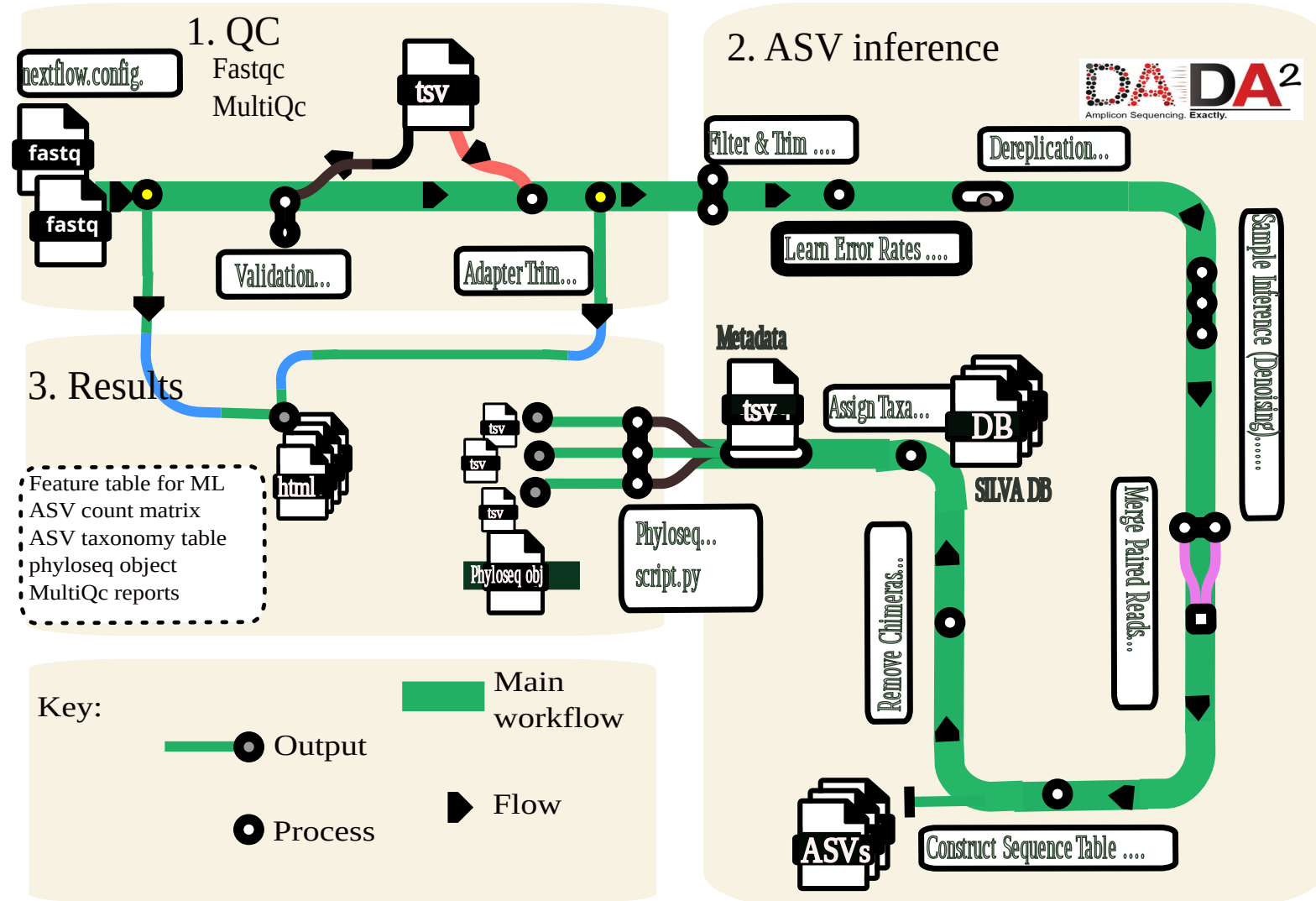
Challenges:

- Broad and rapidly evolving literature.
- Limited computational infrastructure.

How did we handle them?

- Distribution of tasks.
- Developed a nextflow pipeline.

ENEZAflow



Acknowledgements

Project 8:

- Samuel Mwasya
- Linnytiffan Moreka
- Geoffrey Katana
- Samson Otiso

ICIPE/ENEZA-DSI

- Dr. Dan Masiga
- Dr. Kennedy Senagi
- Dr. Jacqueline Wahura
- Rose Marubu
- Peninah Safari
- Trainers

Pwani University

- Prof. Suhaila Hashim
- Dr. Zaneta Kidiavai
- Felix Lisso
- Department of Biochemistry and Biotechnology

